

Runbook — Confluent Platform + Flink on Minikube

End-to-end operational guide for running `confluent-kafka-isotope` locally on **Minikube**: cluster → Kafka/Confluent Platform → Flink → SQL reports → traffic → observe → teardown. Every step maps to a target in the [Makefile](#), which is the source of truth.

This is the **Confluent Platform + Flink (self-managed) on Minikube** path. The Confluent Cloud for Apache Flink (CCAF) path is entirely different — Terraform-driven via `make cc-flink-reports-up`, no Minikube — see [root README §3.3](#).

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1.0 Prerequisites (once per machine)

```
make install-prereqs    # docker, kubectl, minikube, helm, gettext,
gradle, openjdk17
make check-prereqs     # verify they're on PATH
```

Default Minikube sizing (override via env): `MINIKUBE_CPUS=6`, `MINIKUBE_MEM=20480`, `MINIKUBE_DISK=50g`. The node architecture is auto-detected so the right `cp-flink` image (amd64/arm64) is selected.

2.0 Cluster + Confluent Platform

```
make cp-up              # = check-prereqs → minikube-start → operator-
install → cp-deploy
make cp-watch           # watch pods come up (Ctrl+C to exit); or: make
cp-status
```

`cp-up` boots Minikube, installs the CFK (Confluent for Kubernetes) operator, and deploys Kafka (KRaft) + Schema Registry + Connect + ksqlDB + REST Proxy + Control Center into the `confluent` namespace.

`cp-up` deliberately does **not** bring up Flink — run §3.0 Flink separately.

3.0 Flink

```
make cp-flink-up          # cert-manager → operator → MinIO → CMF 2.4 →
env → RBAC → app image
make cmf-status           # (~5 min the first time)
```

This installs cert-manager, the Confluent Flink Kubernetes Operator, **MinIO** (S3-compatible store for CMF artifacts), **CMF 2.4** (with artifact management + the writable environment catalog enabled), creates the `dev-local` Flink environment, applies the Flink RBAC, and builds the custom `cp-flink` 2.1 image (`isotope-cp-flink-sql:local`) that bakes in the Kafka + Avro SQL connectors and the S3 filesystem plugin.

The reports run as a CMF *Application*, not SQL statements. CMF's SQL-statement runtime (`io.confluent.flink.FlinkCompiledPlanExecutor`) ships only in the `cp-flink-sql` image, which exists only at **Flink 1.19** — and two reports are `ProcessTableFunctions`, a **Flink 2.x** feature. So all seven reports deploy as a single Flink 2.1 CMF **Application** (entry point `IsotopeReportsJob`), which runs the same `.fql` and surfaces in CMF / Control Center as a managed application. No raw session cluster is created — if you want one for ad-hoc SQL, deploy it separately with `make flink-deploy` (see §7.0 Observe).

CMF license. The `cp-cmf` image ships with a date-locked trial license. Once it expires, the CMF pod `CrashLoopBackOffs` on startup (`LicenseInitiator` throws) and `make cp-flink-up` fails at the CMF readiness wait. To run past expiry, supply a Confluent license via a secret and point `CMF_LICENSE_SECRET` at it:

```
kubectl create secret generic confluent-license-for-cmf -n confluent \
  --from-file=license.txt=/path/to/license.txt # key MUST be
license.txt
make cp-flink-up CMF_LICENSE_SECRET=confluent-license-for-cmf
```

Export `CMF_LICENSE_SECRET` in your shell to make plain `make cp-flink-up` pick it up.

4.0 Port-forward Kafka

```
make kafka-pf-up          # localhost:30092 → Kafka, localhost:8081 → SR
```

Prereq for everything the host-run gradle app does — `App.java`'s defaults already point at `localhost:30092` / `localhost:8081`. Stop with `make kafka-pf-down`.

5.0 Deploy the 7 Flink reports (as a CMF Application)

```
make cp-flink-reports-up      # builds the app shadow JAR (:ptf:shadowJar) →
pre-creates                  # 4 source + 7 sink topics → uploads the JAR as
a cmf://                      # artifact (MinIO) → deploys the
FlinkApplication → waits RUNNING
```

All seven reports run in **one** Flink 2.1 CMF Application (**isotope-reports**): five pure Flink SQL plus two JAR-backed **ProcessTableFunctions** (**LatencyPercentilesPTF**, **StuckTracePTF**), executed together as a single **StatementSet** by **IsotopeReportsJob**. Sink topics use Apache Flink's **avro-confluent** format — SR-framed Avro, auto-registered on first write — so Control Center renders the rows natively. The application appears in CMF's applications API and Control Center's Flink tab. Drop everything (application + artifact + sink topics) with **make cp-flink-reports-down**.

Optional — fan-in provenance. Off by default. To also run the merge collector and its merge-edge sideband:

```
make cp-flink-reports-up ENABLE_MERGE_PROVENANCE=true
```

That adds two topics (**orders.flink_batched**, **isotope_merge_edge_markers**) and two more **INSERTs** to the same **StatementSet**. The merged records carry a fresh trace, and the edge topic records which parent traces fed each one — one row per contributing trace per window, a material fraction of the merge stage's write volume. See [docs/flink-collector.md §2.4](#). Teardown removes those topics either way.

[OPTIONAL] State-level provenance (§3.6). A second, independent switch:

```
make cp-flink-reports-up ENABLE_STATE_PROVENANCE=true

# both collectors — they answer different questions
make cp-flink-reports-up ENABLE_MERGE_PROVENANCE=true
ENABLE_STATE_PROVENANCE=true
```

That adds one topic (**isotope_state_provenance**) and one **INSERT** to the same **StatementSet**. Each traced order becomes an entity whose three stage records are three versions of it, chained by a **parents** array — **orders.placed** → **orders.enriched** → **orders.fulfilled**. Unlike the merge collector this needs no window: identity is content-addressed, so it works on the updating sources (unbounded **GROUP BY**, joins, dedup) where the windowed path has nothing to key on. **CP only** — see [docs/state-provenance.md §5.0](#) for why CCAF cannot run this statement verbatim. Teardown removes the topic either way.

Watch it fill:

```
kubectl exec -n confluent kafka-0 -- kafka-console-consumer \
  --bootstrap-server localhost:9071 --topic isotope_state_provenance --
  from-beginning --max-messages 5
```

6.0 Drive traffic (required to see report rows)

All report jobs aggregate over `TUMBLE(event_time, INTERVAL '1' MINUTE)` windows, which only emit when the watermark advances past `window_end`. Spread records across **multiple** windows:

```
# 30 records, 5s apart ≈ 2.5 min of event-time → spans 3+ windows
for i in {1..30}; do ./gradlew :app:run --args="place burst-$i" -q; sleep
5; done
```

Or run the pipeline-position stages, each in its own terminal:

```
./gradlew :app:run --args="place hello" # origin → orders.placed
./gradlew :app:run --args="enrich"      # orders.placed →
orders.enriched
./gradlew :app:run --args="fulfill"     # orders.enriched →
orders.fulfilled
./gradlew :app:run --args="ship"        # terminal consume
orders.fulfilled
```

Wait ~90s after the **last** record before checking results — that's the watermark catching up.

`stuck_trace_alerts_1m` only fires for a trace that goes ≥60s of event time without a fresh hop; to exercise it, send one record to `orders.placed`, skip the `enrich/fulfill` hops, and keep sending unrelated records so the watermark crosses `event_time + 60s`.

7.0 Observe

```
make c3-open          # Control Center – report topics render natively
(Avro+SR)
make flink-ui         # Flink job graph / watermarks for the reports
(localhost:8082)
make metrics-up       # optional: Prometheus + Grafana for the stateless
meters
```

The metrics showcase has its own runbook — [k8s/monitoring/README.md](#) (and the meter/PromQL reference in [docs/metrics.md](#)). Stop the background port-forwards with `make c3-stop / make flink-ui-stop / make metrics-down`.

Ad-hoc Flink SQL — needs a session cluster (optional). The reports run as a CMF *Application* (§5.0): one compiled job under `execution.target=kubernetes-application`, which **cannot accept ad-**

hoc SQL. Interactive querying needs a separate **session cluster**, and neither **cp-flink-up** nor **cp-flink-reports-up** creates one. Deploy it first:

```
make flink-deploy      # one-time: deploy the 'flink-basic' session
cluster (~40s)
make flink-sql         # interactive SQL Client on it, report DDL
preloaded
```

flink-deploy applies [k8s/base/flink-cluster-deployment.yaml](#) — a **cp-flink** 2.1 session cluster with 8 task slots, whose init containers fetch the Kafka and Avro-SR connector JARs. **flink-sql** then installs the PTF JAR into the JobManager's `/opt/flink/lib` and preloads the DDL from [scripts/flink/sql/cp/](#), so the session opens ready to query:

```
Flink SQL> SHOW TABLES;           -- 4 source tables + 7 report sinks +
views
Flink SQL> SHOW USER FUNCTIONS;    -- latency_percentiles, stuck_trace_ptf
```

Skip **flink-deploy** and **make flink-sql** stops with a message telling you to run it — it will not fall back to the reports Application pod. **make flink-ui** opens the reports Application by default; point it at this session cluster with **make flink-ui FLINK_UI_TARGET=flink-basic** to watch ad-hoc queries.

The SQL Client catalog is its own. The preloaded DDL lives in that session only, and the running reports Application registers its tables inside the job's own **TableEnvironment** — no shared catalog exists, so **SHOW TABLES** never reflects what the Application is running. Only DDL is preloaded; the report **INSERT INTO** files (`10_...70_`) are deliberately excluded, because the SQL Client rejects DML in an init file. Run them by hand only if you mean to — the Application is already writing to those same sink topics, so you would get duplicate rows, not an error.

8.0 Teardown

Pick the depth:

```
make cp-flink-reports-down  # drop reports / views / functions only
make flink-delete          # drop just the 'flink-basic' ad-hoc SQL
session cluster
make metrics-delete        # remove the Prometheus/Grafana showcase (pods
+ namespace)
make cp-flink-down         # Flink cluster + CMF + operator + cert-manager
(includes flink-delete)
make cp-down               # CP + operator (keeps Minikube running)
make cp-teardown           # everything + stop Minikube
make nuke                  # cp-teardown + minikube-delete + uninstall-
prereqs (factory reset)
```

9.0 Minimal path (no Flink)

To just watch a trace propagate without the Flink SQL reports:

```
make cp-up
make kafka-pf-up
./gradlew :app:run --args="place hello" # then enrich / fulfill / ship
```

Flink (§3.0 Flink) and the reports (§5.0 Deploy the 7 Flink reports (as a CMF Application)) are only needed for the SQL report topics.

10.0 Troubleshooting

- **Pods stuck Pending.** Minikube is under-resourced — raise `MINIKUBE_CPUS` / `MINIKUBE_MEM` and `make minikube-delete && make cp-up`.
- **Flink job submission fails (services forbidden).** The supplemental RBAC didn't apply — `make flink-rbac`.
- **No report rows.** Almost always the watermark — see §6.0 Drive traffic (required to see report rows); traffic must span multiple 1-minute windows and you must wait ~90s after the last record.
- **App can't reach Kafka.** `make kafka-pf-up` isn't running, or the forward died — re-run it and confirm `localhost:30092` / `localhost:8081` are live.
- **`make flink-sql` says no session cluster / `SHOW TABLES`; returns an empty set.** Ad-hoc SQL needs the `flink-basic` session cluster — run `make flink-deploy` first, see §7.0 Observe. Confirm with `kubectl get flinkdeployment -n confluent: isotope-reports` alone means only the reports Application is up. Note that the report tables of the running Application are never visible to the SQL Client regardless — the catalogs are separate.
- **Control Center's Flink tab is blank.** CMF proxy connectivity — `make cmf-proxy-inject` (and `make cmf-proxy-logs` to debug).
- **`make cp-flink-up` times out waiting for the CMF pod / CMF `CrashLoopBackOff`.** Check `kubectl logs -n confluent -l app.kubernetes.io/name=confluent-manager-for-apache-flink`. If you see `Trial license ... expired / LicenseInitiator ... Constructor threw exception`, the image's embedded trial license has expired — supply your own via `CMF_LICENSE_SECRET` — see the CMF license note in §3.0 Flink. Wiping the CMF `PersistentVolumeClaim` does **not** reset it; the expiry is baked into the image.